



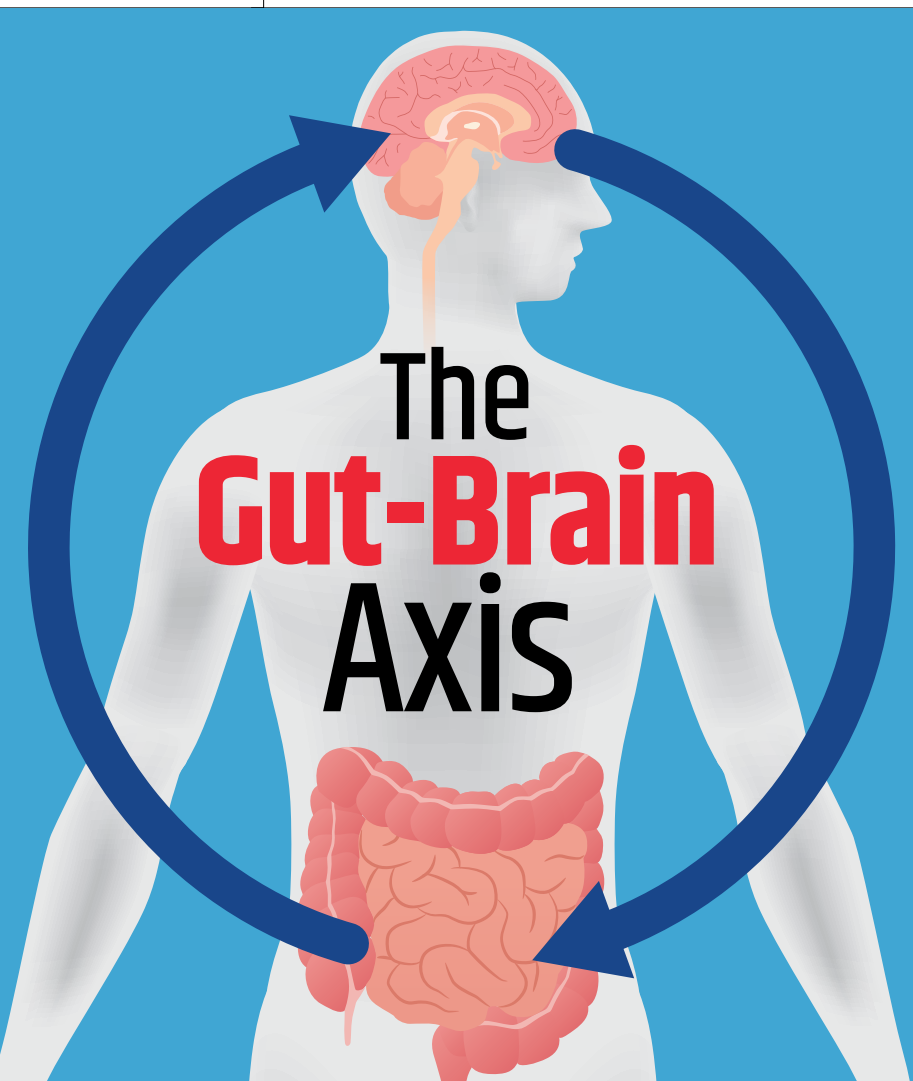
Hydrogen fuel from seawater

Researchers have devised a way to generate hydrogen fuel using solar power, electrodes and saltwater. <https://digit.in/apr19-1>



Study: Eat fewer eggs

Eating more dietary cholesterol, e.g whole eggs causes higher risk of cardiovascular disease. <https://digit.in/apr19-2>



The stomachs and brains in animals are constantly sending little understood biochemical signals back and forth.

Aditya Madanapalle |
aditya@digit.in

T

he enteric nervous system is one of the parts of the autonomic nervous system, with about 500

million neurons that governs the functions of the gastrointestinal tract. For comparison, the spinal cord has only 100 million neurons. The enteric nervous system is referred to as the “second brain”. Till recently, there was

little research into what kind of messages are passed between the neurons in the brain and the neurons in the guts, and the ways in which the body reacts to these signals. Now there are many ways for the guts to talk to the brain. They can release hormones, such as leptin and ghrelin in the bloodstream. These hormones signal hunger or satiation to the central nervous system, a process that takes about ten minutes. The vagus nerve goes from the brainstem to the stomach, and there are enteroendocrine cells lining the guts that send signals directly to the brain

through electrical signals. These signals travel through the body faster than the blink of an eye. The connection of gut neurons to the vagus nerve was discovered only in 2010, by Diego Bohorquez, a neuroscientist at Duke University. The benefit of the pathway is poorly understood, but there are many potential benefits, such as signalling the ingestion of toxins or communicating the caloric value of food to the brain.

THE HUNGRY WORM

The gut to brain connection dates back to the first multicellular organisms that emerged about 600 million years ago. How early in the evolutionary tree the pathway emerged, signifies how important it is for the everyday functioning of all the animals.

In a recent study, researchers from MIT looked into a particular type of cell in an organism called *Caenorhabditis elegans*. This nematode is a favourite in laboratories around the world because of its simple nervous system, made up of only 302 neurons. A known behaviour of *C. elegans* was that it would stop to feed when food was plentiful, and start roaming around when the food was scarce. While the behaviour was understood, the underlying mechanism was not. Steven Flavell, the senior author of the study, explains, “food is something that really motivates this animal, so people have studied this for a long time, but the mechanism of how food ingestion is detected by the nervous system to drive a behavioural change, that had really been missing.” Over a series of experiments, the researchers found out. The guts of the worm, an organ known as the pharynx has neurosecretory-motor (NSM) neurons. These NSM neurons have a single neurite, a sensory tip that extends into the pharynx. The researchers discovered two new ion channels at the end of the neurites, belonging to a family of proteins called acid-sensing ion channels (ASICs). The neurites would sense the presence of bacteria, which is the primary food source of the creatures.